THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

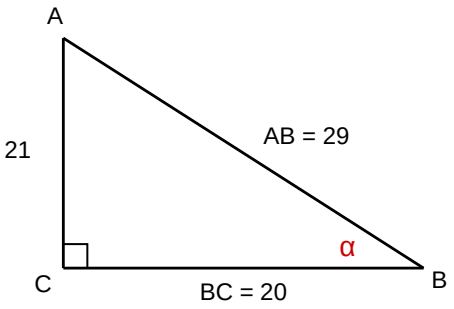
ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 10 · SET 04

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION – A (Multiple Choice Questions · 20 × 1 = 20) — **20 questions · 20 marks**

Q1–Q18 are multiple-choice questions. Q19–Q20: a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option.

1. From the given figure, the value of sin α is



(a) $\frac{20}{29}$ (b) $\frac{21}{20}$ (c) $\frac{21}{29}$ (d) $\frac{20}{21}$

Ans: (c) sin α = AC/AB = 21/29.

2. In ΔABC, D and E are points on AB and AC with DE ∥ BC. If AD = 2 cm, AB = 6 cm and AC = 9 cm, then AE is

(a) 4 cm (b) 6 cm (c) 2 cm (d) 3 cm

Ans: (d) AD/AB = AE/AC ⇒ 2/6 = AE/9 ⇒ AE = 3 cm.

3. The HCF and LCM of 6, 72 and 120 are respectively

(a) 6, 360 (b) 12, 720 (c) 6, 720 (d) 12, 360

Ans: (a) 6 = 2·3; 72 = $2^3 \cdot 3^2$; 120 = $2^3 \cdot 3 \cdot 5$ ⇒ HCF = 2·3 = 6; LCM = $2^3 \cdot 3^2 \cdot 5$ = 360.

4. A natural number is chosen at random from 1 to 60. The probability that it is a multiple of 7 is

(a) $\frac{7}{60}$ (b) $\frac{2}{15}$ (c) $\frac{3}{20}$ (d) $\frac{1}{7}$

Ans: (b) Multiples of 7 up to 60: 7, …, 56 = 8 numbers ⇒ 8/60 = 2/15.

5. Which of the following points CANNOT lie on the line $x + y = 0$?

(a) (0, 0) (b) (2, 2) (c) (3, −3) (d) (−5, 5)

Ans: (b) For (2, 2): 2 + 2 = 4 ≠ 0.

6. (5 + √7)(5 − √7) equals

(a) 32 (b) 12 (c) 18 (d) 25 − 7√7

Ans: (c) 25 − 7 = 18.

7. If the HCF of 65 and 117 is expressible as $65m - 117$, then the value of m is

(a) 2 (b) 1 (c) 3 (d) 4

Ans: (a) HCF = 13 ⇒ 65m = 130 ⇒ m = 2.

8. A person saves ₹ 100 in the first month and ₹ 20 more every following month. The saving in the 8th month is

(a) ₹ 220 (b) ₹ 260 (c) ₹ 200 (d) ₹ 240

Ans: (d) $a_8 = 100 + 7 \times 20 = 240$.

9. For which value of k do $3x + ky = 9$ and $6x + 8y = 18$ have infinitely many solutions?

(a) 3 (b) 4 (c) 6 (d) 8

Ans: (b) $3/6 = k/8 = 9/18 \Rightarrow k = 4$.

10. The value of $\sec^2\theta - \tan^2\theta$ is

(a) 0 (b) −1 (c) $\sin^2\theta$ (d) 1

Ans: (d) Identity $\sec^2\theta - \tan^2\theta = 1$.

11. The point on the x-axis equidistant from A(−2, 0) and B(6, 0) is

(a) (4, 0) (b) (0, 2) (c) (−2, 6) (d) (2, 0)

Ans: (d) Midpoint of AB = (2, 0).

12. Two coins are tossed together. The probability of getting at most one head is

(a) $\frac{1}{4}$ (b) $\frac{1}{2}$ (c) $\frac{3}{4}$ (d) 1

Ans: (c) HH excluded: TT, HT, TH $\Rightarrow$ 3/4.

13. The solution of $x + y = 11$ and $x - y = 3$ is

(a) x = 4, y = 7 (b) x = 8, y = 3 (c) x = 7, y = 4 (d) x = 6, y = 5

Ans: (c) $2x = 14 \Rightarrow x = 7$, $y = 4$.

14. The LCM of two numbers is 2400 and HCF is 20. If one is 400, the other is

(a) 60 (b) 120 (c) 240 (d) 100

Ans: (b) $2400 \times 20/400 = 120$.

15. A die is thrown once. The probability of getting a prime number greater than 2 is

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$ (c) $\frac{2}{3}$ (d) $\frac{1}{3}$

Ans: (d) 3 and 5 $\Rightarrow$ 2/6 = 1/3.

16. If $\sin\theta = \frac{3}{5}$ for an acute angle θ, then $\tan\theta$ equals

(a) $\frac{4}{3}$ (b) $\frac{3}{5}$ (c) $\frac{3}{4}$ (d) $\frac{5}{4}$

Ans: (c) $\cos\theta = 4/5 \Rightarrow \tan\theta = 3/4$.

17. The lines $2x + 3y = 5$ and $4x + 6y = 7$ are

(a) coincident (b) parallel (c) intersecting (d) perpendicular

Ans: (b) $2/4 = 3/6 \neq 5/7 \Rightarrow$ no solution, parallel.

18. The 5th term of the AP −3, −1, 1, 3, … is

(a) 5 (b) 7 (c) 4 (d) 3

Ans: (a) $a_5 = -3 + 4 \times 2 = 5$.

19. **Assertion (A):** The point (−4, 0) lies on the x-axis.
Reason (R): The y-coordinate of every point on the x-axis is zero.

Ans: (a) Both true; R explains A.

20. **Assertion (A):** The sum of two irrational numbers is always irrational.
Reason (R): $\sqrt{2}$ and $-\sqrt{2}$ are irrational.

Ans: (d) A is false: $\sqrt{2} + (-\sqrt{2}) = 0$ is rational. R is true ⇒ (d).

SECTION – B (Very Short Answer Questions · 5 × 2 = 10) — 5 questions · 10 marks

21. Show that 12^n cannot end with the digit 0 for any natural number n.

$12^n = 2^{2n} \cdot 3^n$ has no factor 5 (1); a number ending in 0 is divisible by 5 ⇒ impossible (1).

O R

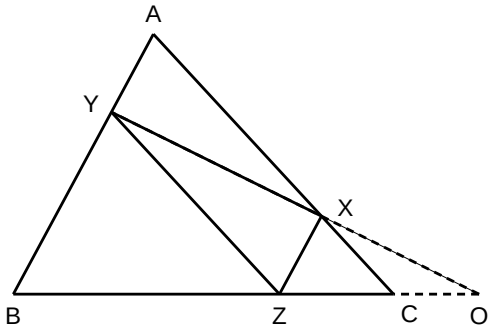
Two runners take 12 min and 18 min to complete a round of a park. If they start together, after how many minutes will they meet again at the start?

LCM(12, 18) = 36 (1) ⇒ 36 minutes (1).

22. A girl 1.5 m tall casts a shadow 2 m long. At the same time a tree casts a shadow 12 m long. Find the height of the tree.

1.5/2 = h/12 (1) ⇒ h = 9 m (1).

23. In the figure, Z is on BC, X on AC and Y on AB. State why the triangle ratio AB/AC = BZ/ZC applies when AZ bisects ∠A, and find BZ if AB = 6, AC = 9 and BC = 10.



Angle bisector theorem: BZ/ZC = AB/AC = 2/3 (1) ⇒ BZ = 4 cm (1).

O R

If ΔABC ~ ΔDEF, AB = 4 cm, DE = 6 cm, EF = 9 cm and FD = 12 cm, find the perimeter of ΔABC.

Scale 4/6 = 2/3 (1); perimeter of DEF = 27 ⇒ ABC = 18 cm (1).

24. A ladder 10 m long reaches a window $5\sqrt{3}$ m above the ground. Find the angle the ladder makes with the ground.

$\sin\theta = 5\sqrt{3}/10 = \sqrt{3}/2$ (1) ⇒ $\theta = 60°$ (1).

25. Find the distance between the points (2, −3) and (−4, 5).

$d = \sqrt{((-6)^2 + 8^2)} = \sqrt{100} = 10$ (2).

SECTION – C (Short Answer Questions · 6 × 3 = 18) — 6 questions · 18 marks

26. Prove that $3 + 2\sqrt{5}$ is irrational, given $\sqrt{5}$ is irrational.

Assume $3 + 2\sqrt{5} = p/q$ (1) ⇒ $\sqrt{5} = (p - 3q)/(2q)$ is rational (1), a contradiction. Hence $3 + 2\sqrt{5}$ is irrational (1).

27. Solve: $3x + 4y = 10$ and $2x - 2y = 2$.

$x - y = 1 \Rightarrow x = y + 1$ (1); $3y + 3 + 4y = 10 \Rightarrow y = 1$ (1), $x = 2$ (1).

28. How many terms of the AP 9, 17, 25, … must be taken so that the sum is 636?

$n/2\,[18 + (n - 1)8] = 636 \Rightarrow 4n^2 + 5n - 636 = 0 \Rightarrow n = 12$ (3).

OR

The 4th term of an AP is 0. Prove that its 25th term is triple its 11th term.

> $a + 3d = 0 \Rightarrow a = -3d$ (1). $a_{25} = a + 24d = 21d$ (1); $a_{11} = a + 10d = 7d \Rightarrow a_{25} = 3a_{11}$ (1).

29. Find the ratio in which the point P(2, y) divides A(−2, 2) and B(3, 7). Also find y.

> $2 = (3k - 2)/(k + 1) \Rightarrow k = 4$ (1½); $y = (7\times4 + 2)/5 = 6$ (1½). Ratio 4 : 1.

OR

Find the area of the triangle with vertices (1, −1), (−4, 6) and (−3, −5).

> ½|1(6+5) + (−4)(−5+1) + (−3)(−1−6)| = ½|11 + 16 + 21| = 24 sq. units (3).

30. If $\cos\theta = \frac{8}{17}$, find the value of sin θ and tan θ.

> sin θ = 15/17 (1½); tan θ = 15/8 (1½).

31. Two dice are thrown together. Find the probability that the sum of the numbers is (a) 8 (b) at least 10 (c) a perfect square.

> (a) 5/36 (1). (b) (4,6),(5,5),(6,4),(5,6),(6,5),(6,6) = 6/36 = 1/6 (1). (c) sums 4 (3 ways) and 9 (4 ways) ⇒ 7/36 (1).

SECTION – D (Long Answer Questions · 4 × 5 = 20) — 4 questions · 20 marks

32. If a line is drawn parallel to one side of a triangle to intersect the other two sides at distinct points, prove that the other two sides are divided in the same ratio.

> Given ΔABC, DE ∥ BC (½); to prove AD/DB = AE/EC (½). Join BE, CD; draw EM ⊥ AB, DN ⊥ AC (1). ar(ADE)/ar(BDE) = AD/DB; ar(ADE)/ar(CED) = AE/EC (1½). ΔBDE and ΔCED on the same base DE, between same parallels ⇒ equal areas (1). Hence AD/DB = AE/EC (½).

33. The sum of the digits of a two-digit number is 9. Nine times the number is twice the number obtained by reversing the digits. Find the number.

> $x + y = 9$ (1); $9(10x + y) = 2(10y + x) \Rightarrow 88x = 11y \Rightarrow y = 8x$ (1½) ⇒ $9x = 9$, $x = 1$, $y = 8$ (1½). Number = 18 (1).

OR

Solve graphically: $x + y = 6$ and $x - y = 2$. Find the area of the triangle formed by the lines and the x-axis.

> Tables (2); graph (1); solution (4, 2) (1); x-intercepts 6 and 2 ⇒ base 4, height 2 ⇒ area 4 sq. units (1).

34. A man on a tower 60 m high observes two cars on the same side of the tower, in a line with its foot, at angles of depression 30° and 60°. Find the distance between the cars. (Take √3 = 1.732)

> Distances from the foot: $60/\tan 30° = 60\sqrt{3}$ and $60/\tan 60° = 20\sqrt{3}$ (2+2). Difference = $40\sqrt{3} = 69.28$ m (1).

35. Prove that $(\sin\theta + \operatorname{cosec}\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$.

> LHS = $\sin^2\theta + \operatorname{cosec}^2\theta + 2 + \cos^2\theta + \sec^2\theta + 2$ (2) = $1 + 4 + (1 + \cot^2\theta) + (1 + \tan^2\theta)$ (2) = $7 + \tan^2\theta + \cot^2\theta$ (1).

OR

If $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, $0° < A + B \le 90°$, $A > B$, find A and B and the value of sin A cos B + cos A sin B.

> $A + B = 60°$, $A - B = 30°$ (2) ⇒ $A = 45°$, $B = 15°$ (2). $\sin(A + B) = \sqrt{3}/2$ (1).

SECTION – E (Case Study Based Questions · 3 × 4 = 12) — 3 questions · 12 marks

36. A school organises a mathematics fair. In a spinner game, a pointer spins on a circle divided into 20 equal sectors numbered 1 to 20.

(i) Find the probability of getting an even number.

> 10/20 = 1/2 (1).

(ii) Find the probability of getting a number divisible by 3.

> 3, 6, 9, 12, 15, 18 ⇒ 6/20 = 3/10 (1).

(iii) Find the probability of getting a prime number greater than 10.

> 11, 13, 17, 19 ⇒ 4/20 = 1/5 (2).

OR

(iii) Find the probability of getting a number that is neither a multiple of 4 nor of 5.

> Multiples of 4: 5; of 5: 4; common: 20 (1) ⇒ 8 numbers ⇒ 12/20 = 3/5 (2).

37. A village is mapped on a grid. The school S is at (2, 3), the hospital H is at (8, 11) and the market M is at (5, 7). Distances are in km.

(i) Find the distance between the school and the hospital.

> SH = √(36 + 64) = 10 km (1).

(ii) Show that the market lies at the midpoint of SH.

> ((2+8)/2, (3+11)/2) = (5, 7) = M (1).

(iii) A new point P lies on SH with SP : PH = 1 : 4. Find its coordinates.

> P = ((1×8 + 4×2)/5, (1×11 + 4×3)/5) = (16/5, 23/5) (2).

OR

(iii) Find the point on the x-axis equidistant from S and H.

> $x^2 - 4x + 4 + 9 = x^2 - 16x + 64 + 121 \Rightarrow 12x = 172 \Rightarrow x = 43/3 \Rightarrow (43/3, 0)$ (2).

38. A library arranges books on a staircase-shaped rack. The bottom shelf holds 40 books and each shelf above holds 3 fewer books than the one below. There are 12 shelves.

(i) Write the first term and the common difference of the AP.

> a = 40, d = −3 (1).

(ii) How many books are on the 12th shelf?

> $a_{12} = 40 - 33 = 7$ (1).

(iii) Find the total number of books on the rack.

> $S_{12} = 6(40 + 7) = 282$ (2).

OR

(iii) Which shelf holds 22 books?

> $40 - 3(n - 1) = 22 \Rightarrow n = 7$ (2).